

Estimating the Term Structure of Indian Sovereign Debt

Cubic Spline vs. Nelson-Siegel Estimation of the G-Sec Yield Curve

Fixed Income Securities — Term 4 Individual Project Report

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1. Executive Summary

This project constructs the zero-coupon term structure of Indian sovereign debt as of August 2, 2026 and compares two standard curve-fitting techniques: the natural cubic spline and the Nelson-Siegel parametric model. The starting dataset, pulled from Bloomberg, contained 1,000 raw price quotes spanning India Government Bonds, Principal Strips and Coupon Strips. After cleaning, filtering for near-par instruments to limit coupon and tax-timing distortion and bootstrapping semi-annual spot rates (supplemented by principal-strip zero rates in the two maturity bands where G-Sec liquidity is thin), a hybrid spot rate curve spanning 0.5 to 29 years was assembled.

The cubic spline, being an interpolant rather than a regression, reproduces every input point exactly ($SSE = RMSE = MAE = 0$). Nelson-Siegel, constrained to three shape parameters and a decay parameter, fits the same data with an RMSE of about 13.7 basis points. The two curves track each other closely through the belly of the curve (roughly 5 to 20 years) and diverge mainly at the long end, where the spline follows a sharp dip-and-jump in the bootstrapped rates that the smoother Nelson-Siegel form cannot reproduce. The choice between the two is therefore less about which model “wins” statistically and more about what the curve will be used for — exact repricing of the underlying instruments versus a smooth, economically interpretable summary of the curve.

2. Data & Cleaning Pipeline

The raw Bloomberg extract (Indian_sovereign.csv) contained 1,000 rows across 12 fields — ask prices, coupons, maturity dates and instrument classifications — for a mix of India Government Bonds, Principal Strips and Coupon Strips. Forty-four rows had missing ask prices and were dropped, leaving 956 usable bonds. Time to maturity was computed for every bond relative to the valuation date of August 2, 2026, as $TTM = (Maturity\ Date - 2026-08-02) / 365.25$.

Two further exclusions were applied before the data was split by instrument type. One bond carried a CALL/SINK optionality flag and was removed, since an embedded option distorts an observed yield relative to a plain default-free zero curve. Six Reserve Bank of India Sovereign Gold Bonds were also removed — these are gold-linked instruments rather than conventional rupee debt and have no place on a nominal sovereign curve. A check on the Country/Region and Currency fields confirmed every remaining bond is an INR-denominated Indian issue, so no cross-border contamination was present.

The cleaned data split into three groups: 105 India Government Bonds (d.gsec), 17 Principal Strips (d.pStrip) and 827 Coupon Strips (d.cStrips). The G-Sec group forms the backbone of the coupon-bond bootstrap; the strips are

used separately — both as standalone zero-coupon observations and, where the G-Sec data leaves gaps, to fill those gaps directly.

A price filter was then applied to the 105 G-Secs, retaining only bonds trading near par (ask price between 90 and 110). Bonds priced far from par carry larger coupon and tax-timing effects that can bias a price-based curve, so restricting to near-par issues is a standard way of limiting that noise before bootstrapping. This filter kept 99 of the 105 G-Secs, a 94.29% retention rate, indicating that most outstanding issues were already trading close to par on the valuation date.

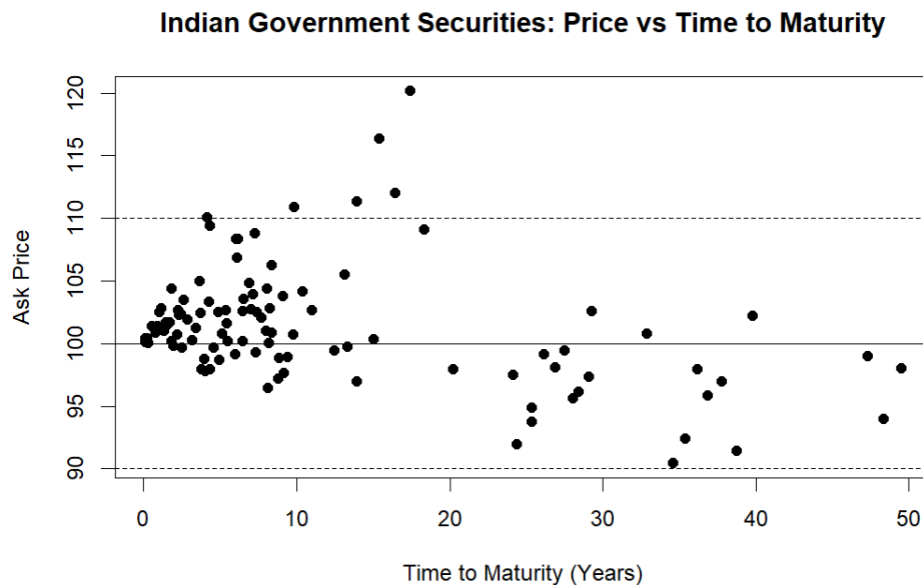


Figure 1. Ask price against time to maturity for the full G-Sec universe, with the 90–110 near-par corridor marked. Bonds outside the band cluster mainly at the very long end and around 15–18 years.

3. Methodology: Hybrid Bootstrapping & Spot Rate Generation

The target grid of the remaining maturity periods for the spot curve runs semi-annually from 0.5 to 29.0 years. Two issues complicate a plain coupon-bond bootstrap on this data. First, near-par issues are not spread evenly across maturities — there are pronounced coverage gaps roughly between 8 and 11 years and again between 15 and 27.5 years. Second, a bootstrap that has to interpolate blindly across gaps that wide, ends up leaning on the interpolation assumption rather than on observed prices, which is exactly what a bootstrap is meant to avoid.

To get around this, spot rates were extracted directly from the 17 Principal Strips using the zero-coupon identity $S = (100/P)^{1/TTM} - 1$ — a strip has no intermediate cash flows to strip out, so its price implies the spot rate directly. Wherever the G-Sec bootstrap would otherwise have had to bridge one of the two large gaps, the strip-implied rates were used instead, interpolated onto the semi-annual grid within those gap regions only.

For every other point on the grid, a semi-annual coupon-bond bootstrap was run. Working through the near-par G-Secs in ascending order of maturity, the code builds each bond's cash flow schedule (coupon/2 every six months, plus principal at maturity) and solves for the single spot rate at that bond's maturity that reprices it back to its observed ask price, given whatever spot rates are already known from earlier iterations or from the strip fill-ins. This search is carried out with R's `uniroot()` over a 0.1%–20% range, applied to $PV(S) - \text{Ask Price}$. Cash flow dates

falling between two already-solved points (for instance a coupon at 3.5 years when the nearest known points are 3.0 and 4.0 years) are filled by linear interpolation between the previous bootstrapped rate and the rate currently being solved for, consistent with standard bootstrap practice. Once solved, a bond's rate is added to the known curve and used in solving the next one.

The result is a hybrid curve — part coupon-bond bootstrap, part strip-implied, part short-gap interpolation — tagged by origin (Bootstrapped G-Sec, Interpolated G-Sec, Principal Strips Extracted) so each component's contribution can be checked visually rather than taken on faith.

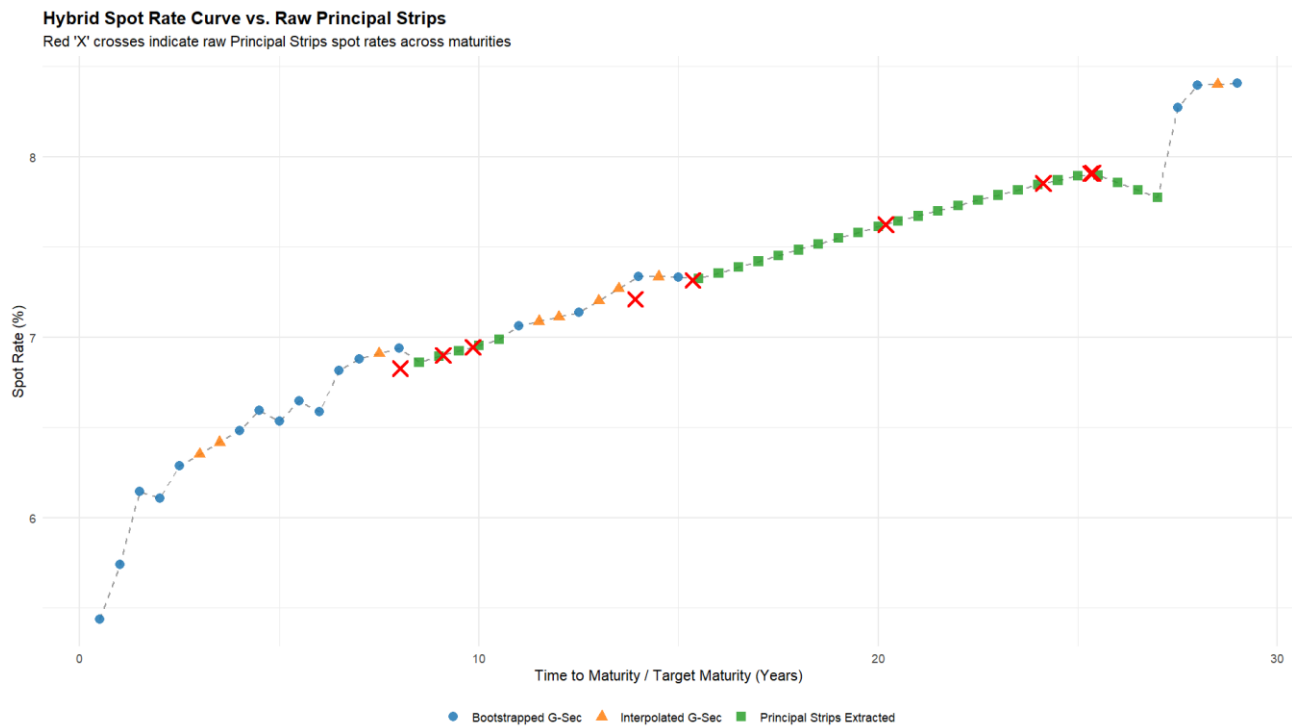


Figure 2. The assembled hybrid spot rate curve, colour- and shape-coded by origin, with raw Principal Strip spot rates overlaid as red crosses. The strips track the hybrid curve closely through both gap regions, confirming the fill-in is consistent with the raw strip data.

4. Yield Curve Fitting Methods

4.1 Natural Cubic Spline

The hybrid curve's (Target Maturity, Spot Rate) pairs were passed to R's `splinefun()` with `method = "natural"`. A natural cubic spline fits a separate cubic polynomial on each interval between consecutive knots,

$$y_i(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3$$

with coefficients chosen so the fitted curve and its first and second derivatives are all continuous across knots (C^2 continuity), and so the second derivative is set to zero at the two endpoints — the “natural” boundary condition, which avoids unconstrained curvature at the edges of the sample. Because a spline has as many effective degrees of freedom as there are knots, it is an interpolant rather than a regression: it will pass through every knot exactly by construction, which is why its fit statistics in Section 5 are all zero.

4.2 Nelson-Siegel Parameterisation

The Nelson-Siegel model represents the spot rate as a three-parameter function of maturity m and a decay parameter τ :

$$y(m) = \beta_0 + \beta_1 \times [(1 - e^{-m/\tau}) / (m/\tau)] + \beta_2 \times [(1 - e^{-m/\tau}) / (m/\tau) - e^{-m/\tau}]$$

β_0 is the level component (the loading is 1 at every maturity, so it shifts the whole curve); β_1 is the slope component (its loading starts at 1 and decays to 0, so it primarily moves the short end relative to the long end); and β_2 is the curvature component (its loading is zero at both $m = 0$ and at long maturities, peaking at an intermediate maturity governed by τ). For a fixed τ , the two loading terms become ordinary regressors and β_0 , β_1 , β_2 can be estimated by OLS — the estimation shortcut proposed by Diebold and Li. τ itself enters the loadings non-linearly, so it was chosen by a grid search over $[0.1, 10.0]$ in steps of 0.1, refitting the OLS regression at each candidate value and keeping the τ that minimises the sum of squared residuals.

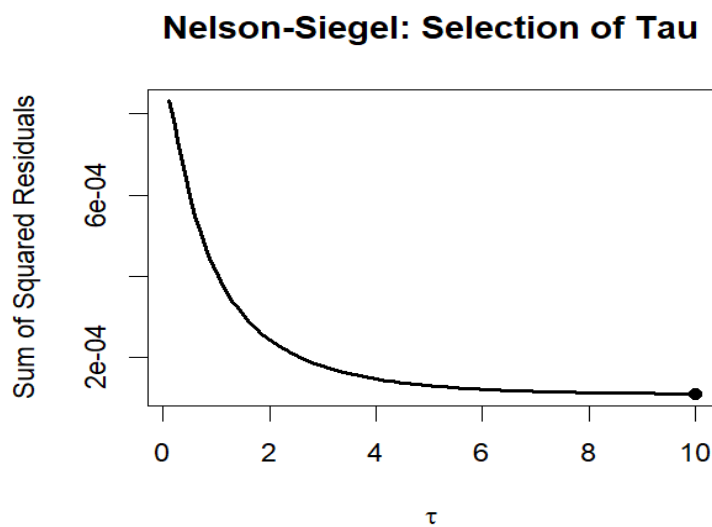


Figure 3. Sum of squared residuals across the τ grid. SSR falls sharply for small τ and flattens out well before $\tau = 10$; the grid search settled on the upper boundary, $\tau = 10.0$, since SSR was still (marginally) declining at that end of the range.

At $\tau = 10.0$, OLS gives $\beta_0 = 0.09420$ (9.42%), $\beta_1 = -0.03581$ (−3.58%) and $\beta_2 = -0.00649$ (−0.65%). The instantaneous short rate implied by the model, $y(0) = \beta_0 + \beta_1$, works out to 5.84%, while the long-run asymptotic rate is simply $\beta_0 = 9.42\%$.

5. Empirical Results & Comparative Analysis

Table 1 summarises the fit of both models against the same 58-point hybrid spot rate curve.

Model	SSE	RMSE	MAE
Cubic Spline	0.0000	0.0000 (0.00%)	0.0000 (0.00%)
Nelson-Siegel	0.0001089	0.001370 (13.7 bps)	0.000970 (9.7 bps)

Table 1. Fit statistics, Cubic Spline vs. Nelson-Siegel.

By construction, the cubic spline dominates on every fit statistic — with as many effective parameters as there are knots, it has no reason not to. The meaningful comparison, then, is not which model fits better but what each error profile buys. Nelson-Siegel’s 13.7 bp RMSE is a genuine estimation residual: three shape parameters plus τ are being asked to summarise close to sixty spot rate observations, so some local wiggle is inevitably left unexplained.

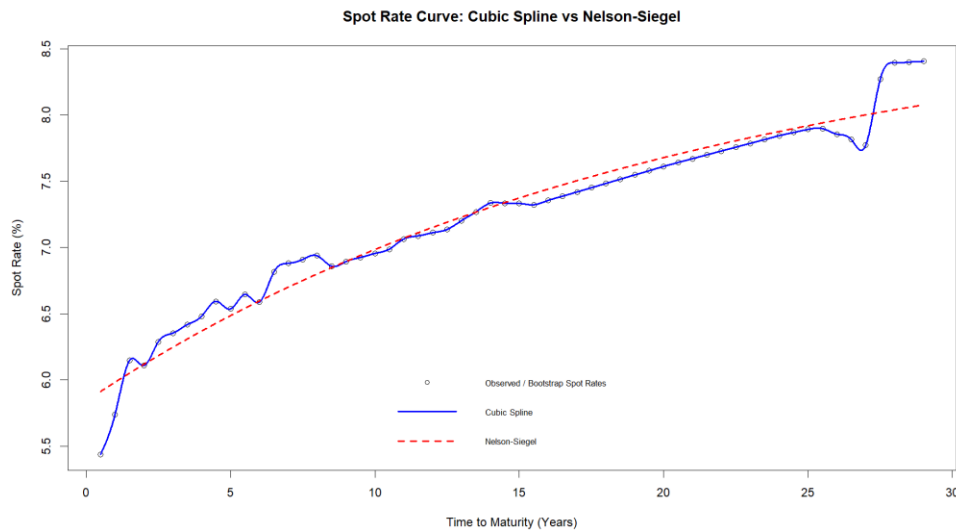


Figure 4. Cubic spline and Nelson-Siegel curves overlaid on the hybrid spot rate observations.

Figure 4 makes the trade-off visible. Through the 5–20 year region the two curves sit almost on top of each other, both tracing the same broadly upward-sloping, gently concave shape. The disagreement concentrates in two places: the short end below about 2 years, where the spline’s local kinks around the 1.5–2 year bootstrapped points are smoothed away by Nelson-Siegel; and the 26–29 year region, where the bootstrapped spot rates dip briefly around 27 years before jumping sharply toward 29 years. The spline follows that dip-and-jump exactly, since it must pass through every point; Nelson-Siegel, which by construction produces at most one hump over the entire curve, has no mechanism to reproduce a second bend near the long end and instead threads a smooth line through it. That single long-end discrepancy accounts for most of the model’s overall RMSE.

In practical terms, the spline is the right tool when the curve needs to reprice the exact instruments used to build it — marking a strips book to market, for instance. Nelson-Siegel is the right tool when the curve needs to behave well outside the range of quoted instruments, stay comparable across dates, or reduce to a handful of economically interpretable numbers.

6. Theoretical Linkages & PCA Factor Mapping

The three Nelson-Siegel parameters are not an arbitrary basis — they map closely onto the level, slope and curvature factors that emerge from a principal component analysis of yield curve changes, a correspondence noted by Litterman and Scheinkman and later formalised by Diebold and Li. Empirically, the first principal component of yield curve movements is close to a parallel shift across all maturities; this is what β_0 captures, since its loading is 1 everywhere and moving it shifts the whole curve together. The second component is a steepening/flattening move, short and long rates moving in opposite directions; this is β_1 ’s role, since its loading

is 1 at the short end and decays toward 0 at long maturities. The third component is a curvature or “hump” move, the belly of the curve shifting relative to both ends; this is exactly what β_2 ’s loading captures, being zero at both extremes and peaking at intermediate maturities near τ .

Read this way, the fitted parameters carry a macro story. $\beta_0 = 9.42\%$, the level factor, is close to what one would expect the market to be pricing in as a long-run nominal anchor once inflation expectations and a term premium are built in — it reflects where policy and inflation are expected to settle over the very long run, not the RBI’s current repo rate. $\beta_1 = -3.58\%$ is negative, meaning the curve is upward sloping (short rates below the long-run level), consistent with a policy stance that is not unusually tight relative to the long-run anchor and with investors demanding a term premium for holding longer paper. β_2 , small and negative at -0.65% , indicates only a mild sag in the belly relative to a straight line between the short and long ends, consistent with Figure 4 showing the curve close to concave-monotonic apart from the two local dislocations already discussed. $\tau = 10$ places the curvature loading’s peak (which occurs near $m \approx 1.76\tau$) around 17–18 years — roughly where the raw price data shows the most pronounced premium pricing (Figure 1) and the steepest local moves in Figure 4.

7. Practical Applications & Code Reproducibility

A clean zero-coupon curve of this kind feeds directly into pricing and risk work: discounting any future INR cash flow, marking G-Sec and strip positions to market, computing forward rates for FRA-style hedges and reading the RBI’s policy stance off the level/slope/curvature decomposition for macro or relative-value views. The cubic spline is the more natural choice for the first two uses, where exact repricing of the underlying instruments matters; Nelson-Siegel is the more natural choice for the latter two, where a small, stable, comparable-across-time parameter set is worth more than a perfect local fit.

Every result in this report can be reproduced from `FIS_project_analysis.R` together with `Indian_sovereign.csv` (downloaded from Bloomberg on August 2, 2026). The script is self-contained: it reads the raw CSV, performs the cleaning and near-par filtering steps of Section 2, runs the hybrid bootstrap of Section 3, fits both models as described in Section 4, and reproduces the `model_comparison` table and every figure referenced above. No random seeds appear anywhere in the pipeline — `uniroot()` is deterministic given its bracket, and the τ grid search is an exhaustive scan — so re-running the script against the same CSV file will reproduce the reported parameter values and error metrics exactly.